--DEMO 5.1 – Функции

```
-- Скалярные функции
SELECT YEAR(SellStartDate) SellStartYear, ProductID, Name
FROM SalesLT.Product
ORDER BY SellStartYear;

SELECT YEAR(SellStartDate) SellStartYear, DATENAME(mm,SellStartDate) SellStartMonth,
       DAY(SellStartDate) SellStartDay, DATENAME(dw, SellStartDate) SellStartWeekday,
          ProductID, Name
FROM SalesLT.Product
ORDER BY SellStartYear;

SELECT DATEDIFF(yy,SellStartDate, GETDATE()) YearsSold, ProductID, Name
FROM SalesLT.Product
ORDER BY ProductID;

SELECT UPPER(Name) AS ProductName
FROM SalesLT.Product;

SELECT CONCAT(FirstName + ' ', LastName) AS FullName
FROM SalesLT.Customer;

SELECT Name, ProductNumber, LEFT(ProductNumber, 2) AS ProductType
FROM SalesLT.Product;

SELECT Name, ProductNumber, LEFT(ProductNumber, 2) AS ProductType,
                            SUBSTRING(ProductNumber,CHARINDEX('-', ProductNumber) + 1, 4) AS
ModelCode,
                                                    SUBSTRING(ProductNumber, LEN(ProductNumber) -
CHARINDEX('-', REVERSE(RIGHT(ProductNumber, 3))) + 2, 2) AS SizeCode
FROM SalesLT.Product;


-- Логические функции
SELECT Name, Size AS NumericSize
FROM SalesLT.Product
WHERE ISNUMERIC(Size) = 1
       -- AND ISDATE(SellEndDate)=1
       -- OR ISNUMERIC(SUBSTRING(ProductNumber,CHARINDEX('-', ProductNumber) + 1, 4))=1

select IIF(5>7, 'true', 'false') as Result

SELECT Name, IIF(ProductCategoryID IN (5,6,7), 'Bike', 'Other') ProductType
FROM SalesLT.Product;

SELECT Name, IIF(ISNUMERIC(Size) = 1, 'Numeric', 'Non-Numeric') SizeType
FROM SalesLT.Product;

SELECT prd.Name AS ProductName, cat.Name AS Category,
      CHOOSE (cat.ParentProductCategoryID, 'Bikes','Components','Clothing','Accessories') AS
ProductType
FROM SalesLT.Product AS prd
JOIN SalesLT.ProductCategory AS cat
ON prd.ProductCategoryID = cat.ProductCategoryID;


-- Оконные функции
SELECT TOP(100) ProductID, Name, ListPrice,
       RANK() OVER(ORDER BY ListPrice DESC) AS RankByPrice
FROM SalesLT.Product AS p
ORDER BY RankByPrice;
```

```sql
SELECT c.Name AS Category, p.Name AS Product, ListPrice,
       RANK() OVER(PARTITION BY c.Name ORDER BY ListPrice DESC) AS RankByPrice
FROM SalesLT.Product AS p
JOIN SalesLT.ProductCategory AS c
ON p.ProductCategoryID = c.ProductcategoryID
ORDER BY Category, RankByPrice;


-- Агрегатные функции (функции группировки)
SELECT COUNT(*) AS Products, COUNT(DISTINCT ProductCategoryID) AS Categories, AVG(ListPrice) AS
AveragePrice
FROM SalesLT.Product;

SELECT COUNT(p.ProductID) BikeModels, AVG(p.ListPrice) AveragePrice
FROM SalesLT.Product AS p
JOIN SalesLT.ProductCategory AS c
ON p.ProductCategoryID = c.ProductCategoryID
WHERE c.Name LIKE '%Bikes';
```

--DEMO 5.2 – Group By

```sql
SELECT Salesperson, COUNT(CustomerID) Customers
FROM SalesLT.Customer
GROUP BY Salesperson
ORDER BY Salesperson;

SELECT c.Name AS Category, COUNT(p.ProductID) AS Products
FROM SalesLT.Product AS p
JOIN SalesLT.ProductCategory AS c
ON p.ProductCategoryID = c.ProductCategoryID
GROUP BY c.Name
ORDER BY Category;

SELECT c.Salesperson, SUM(oh.SubTotal) SalesRevenue
FROM SalesLT.Customer c
JOIN SalesLT.SalesOrderHeader oh
ON c.CustomerID = oh.CustomerID
GROUP BY c.Salesperson
ORDER BY SalesRevenue DESC;

SELECT c.Salesperson, ISNULL(SUM(oh.SubTotal), 0.00) SalesRevenue
FROM SalesLT.Customer c
LEFT JOIN SalesLT.SalesOrderHeader oh
ON c.CustomerID = oh.CustomerID
GROUP BY c.Salesperson
ORDER BY SalesRevenue DESC;

SELECT c.Salesperson, CONCAT(c.FirstName +' ', c.LastName) AS Customer, ISNULL(SUM(oh.SubTotal),
0.00) SalesRevenue
FROM SalesLT.Customer c
LEFT JOIN SalesLT.SalesOrderHeader oh
ON c.CustomerID = oh.CustomerID
GROUP BY c.Salesperson, CONCAT(c.FirstName +' ', c.LastName)
ORDER BY SalesRevenue DESC, Customer;
```

--DEMO 5.3 – Having

```sql
-- Попытка найти менеджеров с более чем 100 клиентами (ошибка)
SELECT Salesperson, COUNT(CustomerID) Customers
FROM SalesLT.Customer
```

```
WHERE COUNT(CustomerID) > 100
GROUP BY Salesperson
ORDER BY Salesperson;

-- Необходимо использовать предложение HAVING для фильтрации, основанной на агрегации
SELECT Salesperson, COUNT(CustomerID) Customers
FROM SalesLT.Customer
GROUP BY Salesperson
HAVING COUNT(CustomerID) > 100
ORDER BY Salesperson;
```